23. Which one of the following lakes of West Africa has become dry and turned into a desert?
(a) Lake Victoria (b) Lake Faguibine
(c) Lake Oguta (d) Lake Volta

Ans. (b) : Lake Faguibine in northern Mali started to disappear after catastrophic droughts in the 1970s. Over several years, droughts in the 1970s dried up the lake.

24. Gandikota canyon of South India was created by which one of the following rivers?
(a) Cauvery (b) Manjira
(c) Pennar (d) Tungabhadra

Ans. (c): Gandikota canyon of South India was created by The river Pennar. The swift-flowing waters of River Pennar cut through the granite rocks near the village of Gandikota, naturally eroding the rock and forming Gandikota Canyon.

25. Consider the following pairs:

Peak		Mountains
1. Namcha Barwa	-	Garhwal Himalaya
2. Nanda Devi	-	Kumaon Himalaya
3. Nokrek	-	Sikkim Himalaya

Which of the pairs given above is/are correctly matched?
(a) 1 and 2 (b) 2 only (c) 1 and 3 (d) 3 only

Ans. (b): Namcha Barwa is a mountain in Arunachal Pradesh and not in Garhwal Himalaya. It is considered as the Eastern point of the Himalayas. Nanda Devi peak is in the state of Uttarakhand. It is a part of the Garhwal Himalayas. The Nokrek is located in the northeast of India on the Tura Range, which forms part of the Meghalaya Plateau. Nokrek is the highest peak of the Garo hills.

26. The term "Levant" often heard in the news roughly corresponds to which of the following regions?
(a) Region along the eastern Mediterranean shores
(b) Region along North African shores stretching from Egypt to Morocco
(c) Region along Persian Gulf and Horn of Africa
(d) The entire coastal areas of Mediterranean Sea

Ans. (a) : The Levant is an approximate historical geographical term referring to a large area in the Eastern Mediterranean region of Western Asia, roughly corresponding to modern-day Israel, Jordan, Lebanon, Syria, and certain adjacent areas.

27. Consider the following countries:
1. Azerbaijan 2. Kyrgyzstan
3. Tajikistan 4. Turkmenistan
5. Uzbekistan

Which of the above have borders with Afghanistan?
(a) 1, 2 and 5 only (b) 1, 2, 3 and 4 only
(c) 3, 4 and 5 only (d) 1, 2, 3, 4 and 5

Ans. (c): Afghanistan is bounded to the east and south by Pakistan (including those areas of Pak occupied Kashmir), to the west by Iran, and to the north by the Central Asian states of Turkmenistan, Uzbekistan, and Tajikistan.

28. With reference to India, consider the following statements:
1. Monazite is a source of rare earths.
2. Monazite contains thorium.
3. Monazite occurs naturally in the entire Indian coastal sands in India.
4. In India, Government bodies only can process or export monazite.

Which of the statements given above are correct?
(a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 3 and 4 only (d) 1, 2, 3 and 4

Ans. (b): Though monazite is found in most coastal areas of India. It is questionable for monazite to be found along entire coast. The main mines are found along the coasts of southern India in Kerala, Tamil Nadu and in Orissa. Rests of the statements are correct.

29. In the northern hemisphere, the longest day of the year normally occurs in the:
(a) First half of the month of June
(b) Second half of the month of June
(c) First half of the month of July
(d) Second half of the month of July

Ans. (b): The longest day of the year for those living north of the Equator is June 21. This day is referred to as the summer solstice. It occurs when the sun is directly over the Tropic of Cancer. Hence the correct answer to the question would be 2nd half of the month of June.

30. Consider the following pairs:

Wetland/Lake		Locations
1. Hokera Wetland	-	Punjab
2. Renuka Wetland	-	Himachal Pradesh
3. Rudrasagar Lake	-	Tripura
4. Sasthamkotta Lake	-	Tamil Nadu

How many pairs given above are correctly matched?
(a) Only one pair (b) Only two pairs
(c) Only three pairs (d) All four pairs

Ans. (b) : The Hokera wetland is a Ramsar site at Zainakote near Srinagar in Jammu and Kashmir. Renuka wetland is in the Sirmaur district of Himachal Pradesh. Rudrasagar Lake, also known as Rudijala is a famous lake situated in the Melaghar Block under Sonamura Sub-Division in the West Tripura District in Tripura. Sasthamkotta lake also known as the Queen of Lakes is the largest freshwater lake in Kerala. Hence only two pairs are correctly matched.

31. Consider the following :
1. Aarogya Setu 2. CoWIN
3. DigiLocker 4. DIKSHA

Which of the above are built on top of open-source digital platforms?
(a) 1 and 2 only (b) 2, 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

Ans. (d): Open source software is software with source code that anyone can inspect, modify, and enhance. Digilocker, Diksha, Aarogya Setu, the Covid-19 vaccination platform CoWIN have been built on top of open-source digital platforms.

32. With reference to Web 3.0, consider the following statements:
1. Web 3.0 technology enables people to control their own data.
2. In Web 3.0 world, there can be blockchain based social networks.
3. Web 3.0 is operated by users collectively rather than a corporation.

Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : Web 3.0 is the third generation web technology. Web 3.0 will have a strong emphasis on decentralized applications and make extensive use of blockchain-based technologies. An emerging governance ideal within the blockchain and Web 3.0 community is the concept of a Decentralized Autonomous Organization (DAO). Instead of having a central authority that governs the operations of a platform, with a DAO, Web 3.0 technologies and communities provide a form of self-governance in an attempted decentralized approach. Presently, only centralized repositories are the ones that own user data and profit from it. In Web 3.0, users can own and be properly compensated for their time and data. In Web 3.0 people will control their own data.

33. With reference to "Software as a Service (SaaS)," consider the following statements:
1. SasS buyers can customise the user interface and can change data fields.
2. SaaS users can access their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d): Software as a service (SaaS) is a way of delivering applications over the Internet as a service. Instead of installing and maintaining software, user can simply access it via the Internet, freeing themselves from complex software and hardware management. Examples of popular SaaS providers include: BigCommerce, Google Workspace, Salesforce, Dropbox, Outlook, Hotmail and Yahoo! Mail etc. Also SasS buyers can customize the user interface and can change data fields and can access their data through their mobile devices. Therefore all the given statements are correct.

34. Which one of the following statements best reflects the idea behind the "Fractional Orbital Bombardment System" often talked about in media?
(a) A hypersonic missile is launched into space to counter the asteroid approaching the Earth and explode it in space.
(b) A spacecraft lands on another planet after making several orbital motions.
(c) A missile is put into stable orbit around the Earth and deorbits over a target on the Earth.
(d) A spacecraft moves along a comet with the same speed and places a probe on its surface.

Ans. (c): A Fractional Orbital Bombardment System (FOBS) is a warhead delivery system that uses a low earth orbit towards its target destination. Just before reaching the target, it de-orbits through a retrograde engine burn. It was developed in 1960s by Soviet Union.

35. Which one of the following is the context in which the term "qubit" is mentioned?
(a) Cloud Services
(b) Quantum Computing
(c) Visible Light Communication Technologies
(d) Wireless Communication Technologies

Ans. (b): A "Qubit" is a quantum bit, the counterpart in quantum computing to the binary digit or bit of classical computing. Just as a bit is the basic unit of information in a classical computer, a qubit is the basic unit of information in a quantum computer.

36. Consider the following communication technologies :
1. Closed-circuit Television
2. Radio Frequency Identification
3. Wireless Local Area Network

Which of the above are considered Short-Range devices/ Technologies?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1,2 and 3

Ans. (d) : Short Range Devices (SRD) are radio devices that offer a low risk of interference with other radio services, usually because their transmitting power, and their low range. The definition 'Short Range Device' may be applied to many different types of wireless equipment, including various forms of:
- Access control (including door and gate openers)
- Alarms and movement detectors
- Closed-circuit television (CCTV)
- Cordless audio devices, including wireless microphones
- Wireless Local Area Networks
- Remote control
- Radio frequency identification (RFID)
- Telemetry etc.

37. Consider the following statements:
1. Biofilms can form on medical implants within human tissues.
2. Biofilms can form on food and food processing surfaces.
3. Biofilms can exhibit antibiotic resistance.

Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : A biofilm is an assemblage of surface-associated microbial cells that is enclosed in an extracellular polymeric substance matrix. Biofilms can form on medical implants within human tissues. It can form on food and food processing surfaces and can exhibit antibiotic resistance. The biofilm provides a favorable environment for the microorganisms.

38. Consider the following statements in respect of probiotics:
1. Probiotics are made of both bacteria and yeast.
2. The organisms in probiotics are found in foods we ingest but they do not naturally occur in our gut.
3. Probiotics help in the digestion of milk sugars.

Which of the statements given above is/are correct?
(a) 1 only (b) 2 only (c) 1 and 3 (d) 2 and 3

Ans. (c) : Probiotics are defined as live micro-organisms which confer a health benefit on the host. Although most probiotics are bacteria, one strain of yeast, Saccharomyces boulardii, has been found to be an effective probiotic in double-blind clinical studies. Probiotics live naturally in our body. L. acidophilus is a probiotic bacteria that naturally occurs in the human gut. Yogurt is a well-known food source of probiotics. Certain strains of bacteria in yogurt have ß-D-galactosidase, which is an enzyme that helps to break down lactose in dairy products into the sugars glucose and galactose. Thus Probiotics help in the digestion of milk sugars.

39. In the context of vaccines manufactured to prevent COVID-19 pandemic, consider the following statements:
1. The Serum Institute of India produced COVID-19 vaccine named Covisheild using mRNA platform.